

Total No. of Questions : 10]

SEAT No :

P 3125

[5154]-691

[Total No. of Pages :2

B.E.(Information Technology)
INFORMATION AND CYBER SECURITY
(2012 Course) (414453)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Answers Question 1 or 2, 3 or 4, 5 or 6, 7 or 8 and 9 or 10.*
- 2) *Neat diagrams must be drawn whenever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data, if necessary.*

- Q1)** a) Compute the inverse of 17 in mod 23 arithmetic. Show steps clearly. [6]
b) State Euler's theorem. [4]

OR

- Q2)** a) Show with proper working that 13 is a primitive root of 19. [6]
b) In Diffie-Hellman key exchange between two parties A and B where A picks his secret as 9 and B picks his secret as 6. Apply 13 as the primitive root of 19, for this Diffie-Hellman exchange and show the shared secret. Show the math working steps clearly. [4]

- Q3)** a) What do you mean by cryptanalysis. Mention the applications of public key cryptography. [6]
b) List out the problems of one time pad. [4]

OR

- Q4)** a) Write down the purpose of S-box in DES. [6]
b) Give the types of attacks with examples. [4]

- Q5)** Consider the following threats to web security and describe how each is countered by particular feature of SSL. [16]

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|-------------------------|-------------------------------|
| a) Brute force attacks. | b) Known plaintext attacks. |
| c) Replay attacks. | d) Man-in-the-middle attacks. |
| e) Password sniffing. | f) IP spoofing. |
| g) IP hijacking. | h) SYN flooding. |

OR

P.T.O.

- Q6)** a) What is the difference between tunnel and transport mode in IPSEC And how does it defend replay attacks. [8]
b) What protocols comprise SSL? What is the difference between SSL connection and SSL session. [8]

- Q7)** a) What is the statistical anomaly detection and rule based intrusion detection and explain the differences between them. [8]
b) Consider a machine M1, hosting a https-based public website www.tech.net is it possible for an intruder with no access to M1 to setup a fake website with the same URL www.tech.net in his machine M2 and go unnoticed? [8]

OR

- Q8)** a) Explain in brief : trap doors, trojan horses, worms and zombies. [8]
b) With a neat diagram explain the process of digital signing and digital verification. [8]

- Q9)** Write notes on : [18]
a) Cyberstalking
b) Cybercrime and cloud computing
c) Phishing

OR

- Q10)** Write notes on : [18]
a) Social engineering attacks
b) ITA 2000
c) DoS and DDos attacks

